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Classical Mechanics & Relativity

We're going to spend the next ~ 2 weeks building up all the pieces we need for doing calculations in particle physics.

In particular, we'll ~~want~~ discuss

- Relativity (since the universe is fundamentally relativistic, & particle physics involves relativistic particles)
- Quantum Mechanical Perturbation Theory
→ the "Fermi's Golden Rule"
(how to calculate what you see in a particle physics experiment)

First, we're going to quickly review the Lagrangian & Hamiltonian formulations of classical mechanics

Lagrangian: the starting point of ^{perturbative} ~~QFTs~~ QFTs, & the language of particle physics.

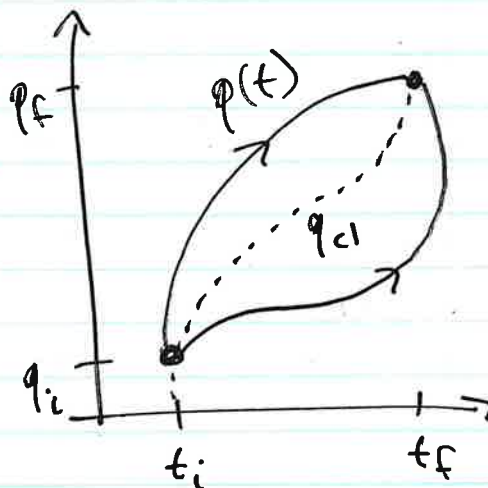
Hamiltonian: the natural path to Quantum Mechanics.

Lagrangian Mechanics

We wish to ~~describe~~^{study} a physical system with one "degree of freedom", described by a "generalized coordinate" $q(t)$.

Suppose the particle (dof) is at q_i at $t = t_i$ and q_f at $t = t_f$.

In principle, there are many different "paths" or "trajectories" the particle could take.



How do we determine the actual trajectory?
(the "classical trajectory", q_{cl})

→ Principle of least action

Lagrangian: function of the generalized coordinate and the corresponding "generalized velocity":

$$L \equiv L[q(t), \dot{q}(t)]$$

From here, we define the action

$$S = \int_{t_1}^{t_2} L(\bar{q}(t), \dot{q}(t)) dt$$

principle of least action: the classical trajectory is the one that minimizes (or more generally, extremizes) the action.

i.e. we want the trajectory that yields: $\delta S = 0$

More explicitly, consider arbitrary variations of the trajectory, $\delta q(t)$ that leave the endpoints fixed:

$$\delta q_i = \delta q(t_i) = 0, \quad \delta q_f = \delta q(t_f) = 0$$

Then:

$$0 = \delta S = \int_{t_1}^{t_2} \delta L(\bar{q}(t), \dot{q}(t)) dt$$

$$\stackrel{\text{(chain rule)}}{=} \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt$$

(4)

Now, $\delta \dot{q} = \delta \left(\frac{dq}{dt} \right) = \frac{d}{dt} (\delta q)$

Integrating by parts on the second term, we find:

$$0 = \delta S = \underbrace{\frac{\partial L}{\partial \dot{q}} \delta q}_{\text{vanishes since } \delta q_i = \delta q_f = 0} \Big|_{t_i}^{t_f} + \int_{t_i}^{t_f} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right] \delta q dt$$

↑
arbitrary

$\Rightarrow$ The solution must satisfy:

$$\left\{ \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right\}_{q_{cl}(t)} = 0 \quad t_i \leq t \leq t_f$$

This is the Euler-Lagrange equation that determines the classical trajectory.

In many physical systems, the Lagrangian is written as the difference of the kinetic and potential energies:

$$L = T - V, \quad T = \frac{1}{2} m \dot{q}^2, \quad V \equiv V(q)$$

(5)

Plugging this into the E-L equations:

$$\frac{\partial L}{\partial q} = \frac{\partial V}{\partial q} \quad , \quad \frac{\partial L}{\partial \dot{q}} = \frac{\partial T}{\partial \dot{q}} = m\dot{q}$$

$$\Rightarrow \delta S = 0 \quad \Leftrightarrow \quad m\ddot{q} = -\frac{\partial V}{\partial q}$$

This is just Newton's second law, so these formulations of classical mechanics are equivalent.

The Lagrangian formalism is powerful because it generalizes easily to systems with multiple (or infinite!) degrees of freedom, with multiple (infinite) generalized coordinates.

This is eventually what we'll want for fields.

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Hamiltonian Formulation

In the Lagrangian formalism, we used q and $\dot{q}$ as independent variables.

Let's define the associated "generalized momentum"

$$p = \frac{\partial L}{\partial \dot{q}}$$

For systems w/ $L = T - V$, $T = \frac{1}{2} m \dot{q}^2$:

$$p = m \dot{q} \quad (\text{"mass times velocity"})$$

Hamiltonian formalism: take q, p as the dynamical variables.

Instead of $L(q, \dot{q})$ we'll define a Hamiltonian:
 $H = H(q, p)$.

The Hamiltonian is obtained from the Lagrangian via a "Legendre transformation".

$$H(q, p) = \dot{q} p - L(q, \dot{q})$$

where it is understood that $\dot{q}$ on the RHS is written in terms of q & p .

(7)

Now, treating q, p as independent variables:
consider

$$\begin{aligned}\frac{\partial H}{\partial p} &= \dot{q} + p \frac{\partial \dot{q}}{\partial p} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p} \\ &= \dot{q} \quad \text{''' } p\end{aligned}$$

and:

$$\begin{aligned}\frac{\partial H}{\partial q} &= p \frac{\partial \dot{q}}{\partial q} - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial q} \\ &= -\frac{\partial L}{\partial q} \quad \text{(E-L eqn.)} \\ &= -\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = -\dot{p}\end{aligned}$$

So we arrive at Hamilton's equations:

$$\frac{\partial H}{\partial p} = \dot{q}, \quad -\frac{\partial H}{\partial q} = \dot{p}$$

→ swapped E-L equations for two first order equations.

(8)

For Lagrangians of the form $L = T - V$, $T = \frac{1}{2} m \dot{q}^2$ and $V = V(q)$, we have:

$$p = \frac{\partial L}{\partial \dot{q}} = m \dot{q} \Rightarrow \dot{q} = p/m$$

Then,

$$\begin{aligned} H &= p \dot{q} - L(q, \dot{q}) \\ &= p \dot{q} - \left[\frac{1}{2} m \dot{q}^2 - V(q) \right] \\ &= p \left(\frac{p}{m} \right) - \frac{1}{2} m \left(\frac{p}{m} \right)^2 + V(q) \end{aligned}$$

or,

$$H = \frac{p^2}{2m} + V(q) = T + V$$

→ The Hamiltonian is associated w/ the mechanical energy of the system.

One can go further and define Poisson brackets, etc., as a route to the Hamilton's equations.

See e.g., Landau & Lifschitz for a very concise treatment.

Before we switch to relativity:

Warm-up: Rotations in Euclidean space

Intuition: a rotation of an ordinary vector leaves its length invariant.

A rotation in 3-dimensions can be represented by a 3×3 matrix R , acting on an arbitrary vector, $\vec{x}$, which transforms (rotates) to the vector $\vec{x}'$.

$$\vec{x}' = R \vec{x}$$

(invariant!)

That its length is preserved is the statement:

$$\vec{x}' \cdot \vec{x}' = (R \vec{x}) \cdot (R \vec{x}) = \vec{x} \cdot \vec{x}$$

We can write this in different ways, e.g. as a column vector multiplying a row vector:

$$\vec{x} \cdot \vec{y} \equiv \vec{x}^T \vec{y} = \vec{x}^T (\mathbb{1}) \vec{y}$$

(T superscript denotes transpose)

w/ $\mathbb{1} = \text{diag}(1, 1, 1)$ is the identity matrix.

Thus, we can write

$$\begin{aligned}\vec{x}' \cdot \vec{x}' &= (\mathbf{R} \vec{x})^T \mathbf{1} (\mathbf{R} \vec{x}) \\ &= \vec{x}^T \mathbf{R}^T \mathbf{1} \mathbf{R} \vec{x} = \vec{x}^T \mathbf{1} \vec{x}\end{aligned}$$

This is true if the matrix $\mathbf{R}$ is orthogonal

$$\mathbf{R}^T \mathbf{1} \mathbf{R} = \mathbf{R}^T \mathbf{R} = \mathbf{1}.$$

We can also write these expressions in terms of the components x_i of the vector, and R_{ij} of the matrices, with $i = 1, 2, 3$.

For example,

$$\begin{aligned}\sum_k x'_k x'_k &= \sum_i \sum_j \sum_k x_i (\mathbf{R}^T)_{ik} R_{kj} x_j \\ &= \sum_i \sum_j x_i \delta_{ij} x_j \quad \left(\begin{array}{l} \text{since} \\ \sum_k (\mathbf{R}^T)_{ik} R_{kj} = \delta_{ij} \end{array} \right) \\ &= \sum_i x_i x_i \quad \left(\sum_j A_j \delta_{ij} = A_i \right)\end{aligned}$$

I will frequently use

Einstein summation convention in which the $\sum$ symbol is dropped, and repeated indices are understood to be summed.

(11)

In these expressions we have emphasized the (usually implicit) δ_{ij} — it lets us take two (of the same, in this case) vectors and returns a distance.

The Kronecker δ -function is the metric tensor for Euclidean space.

Lorentz transformations & Minkowski spacetime

In special relativity, we treat space and time on a (more) equal footing, and work in 4-dimensional spacetime or Minkowski space.

Define a 4-vector that groups spatial/time coords'.

$$x^M = (x^0, x^1, x^2, x^3) = (ct, \vec{x})$$

Greek indices (from the middle of the alphabet)

$\mu, \nu, \dots$ take values 0, 1, 2, 3.

We'll try to reserve Latin indices ($i, j, \dots$) to take values 1, 2, 3 for the spatial components.

A point x^μ defines an "event" in spacetime.

In analogy to our discussion of rotations, we can define a scalar product of four-vectors x^μ, y^μ

$$\begin{aligned} x \cdot y &\equiv x^\mu g_{\mu\nu} y^\nu \\ &= \sum_{\mu,\nu} x^\mu g_{\mu\nu} y^\nu = g_{\mu\nu} x^\mu y^\nu \end{aligned}$$

Where I have introduced the Minkowski Metric

$$g_{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

(This is the "mostly minus" or "West coast" or "particle physics" (as opposed to relativists) convention)

Writing things out term-by-term,

$$\begin{aligned} x \cdot y &= x^0 y^0 - (x^1 y^1 + x^2 y^2 + x^3 y^3) \\ &= x^0 y^0 - \vec{x} \cdot \vec{y} \end{aligned}$$

In contrast to Euclidean space, it's useful in Minkowski space to distinguish between "contravariant" vectors with upper indices and "covariant" vectors with lower indices.

We can always go from one to another by contracting with the metric.

$$x_m \equiv g_{mv} x^v$$

Component-wise, $x_m = (x^0, -x^1, -x^2, -x^3)$

The metric g_{mv} has an inverse g^{mv} which satisfies

$$g g^{-1} = \mathbb{1}$$

or,

$$g_{mv} g^{vp} = \delta_m^p$$

In flat space of course, the components are the same:

$$g^{mv} = \text{diag}(1, -1, -1, -1)$$

We then have a bunch of relations like:

$$x \cdot y = g_{\mu\nu} x^\mu y^\nu = x_\mu y^\mu = x^\mu y_\mu = g^{\mu\nu} x_\mu y_\nu$$

Now we can define more general Lorentz tensors, with a mix of upper & lower indices.

For instance, we might have an object $T^{\mu\nu}_{\rho\sigma}$ which satisfies

$$T^{\mu\nu\lambda}_{\sigma} = g^{\lambda\rho} T^{\mu\nu}_{\rho\sigma}$$

note the order here, as well as position matters!

(e.g., don't write $T^{\mu\nu\lambda}_{\sigma}$)
unless it has some special symmetry properties

It's often useful in manipulations to use our freedom to relabel dummy indices. — if an index is summed over all its components, it doesn't matter what we call it.

As an example, consider a symmetric tensor $S_{\mu\nu}$ satisfying $S_{\mu\nu} = +S_{\nu\mu}$ and an anti-symmetric tensor $A^{\mu\nu} = -A^{\nu\mu}$.

Then,

$$\begin{aligned}
 S_{\mu\nu} A^{\mu\nu} &= \frac{1}{2} S_{\mu\nu} A^{\mu\nu} + \frac{1}{2} S_{\mu\nu} A^{\mu\nu} \\
 &= \frac{1}{2} S_{\mu\nu} A^{\mu\nu} + \frac{1}{2} (S_{\nu\mu}) (-A^{\nu\mu}) \\
 &\quad \text{(using symm. / anti-symm.)} \\
 &= \frac{1}{2} S_{\mu\nu} A^{\mu\nu} - \frac{1}{2} S_{\mu\nu} A^{\mu\nu} \\
 &\quad \text{(just relabelling } \mu \leftrightarrow \nu \text{)} \\
 &= 0!
 \end{aligned}$$

We'll perform such manipulations repeatedly in the context of relativistic QFT.

Just as rotations preserve the length of a Euclidean vector, we can define Lorentz transformations as "rotations" in Minkowski spacetime, which preserve the scalar product.

In matrix form:

$$x' = \Lambda x$$

or in components,

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$$

note the index structure here!

These are defined to leave the scalar product of four-vectors invariant:

$$\begin{aligned} x' \cdot y' &= (x')^T g y' = x^T \Lambda^T g \Lambda y \\ &= x^T g y \end{aligned}$$

the metric

The matrix Λ must then satisfy:

$$\Lambda^T g \Lambda = g$$

This is the analogue of the orthogonality relation,
 $R^T \mathbb{1} R = \mathbb{1}$.

Note that the orthogonality condition implies that g is invariant under ~~metre~~ Lorentz transformations. In components:

$$\Lambda^T g \Lambda = g \Rightarrow (\Lambda^T)_\rho{}^m g_{mn} \Lambda^\nu{}_\sigma = g_{\rho\sigma}$$

$$\text{or, } g_{mn} \Lambda^m{}_\rho \Lambda^\nu{}_\sigma = g_{\rho\sigma} \quad (*)$$

Note that we've used $\Lambda^m{}_\rho = (\Lambda^T)_\rho{}^m$ and reversed the index structure.

This is done so that matrix multiplication follows naturally from adjacent up/down indices.

Any Lorentz transformation Λ has an inverse transformation, Λ^{-1} .

A convenient way to derive its form is to multiply $(*)$ above by the inverse metric:

$$g^{\rho\lambda} \underbrace{[g_{mn} \Lambda^m{}_\rho \Lambda^\nu{}_\sigma]} = g^{\rho\lambda} g_{\rho\sigma} = \delta^\lambda{}_\sigma$$

this quantity $\times \Lambda^\nu{}_\sigma$ is the identity, so by definition it is the inverse matrix:

$$(\Lambda^{-1})^\lambda{}_\nu \Lambda^\nu{}_\sigma = \delta^\lambda{}_\sigma$$

where

$$(\Lambda^{-1})^\gamma_\nu \equiv \Lambda_\nu{}^\gamma = g_{\nu\mu} g^{\mu\rho} \Lambda^\mu_\rho$$



used symmetry of the metric tensor.

Here I've introduced a convenient shorthand, where Λ with the "wrong" index structure in components represents the components of the inverse metric.

Note that Λ^{-1} has the same structure as Λ^T

Note that a (covariant) vector ~~transforms~~ w/ a lower index transforms by the action of Λ^{-1} , rather than Λ .

$$\begin{aligned} x'_\nu &= g_{\nu\mu} x'^\mu = g_{\nu\mu} (\Lambda^\mu_\rho x^\rho) \\ &= g_{\nu\mu} \Lambda^\mu_\rho g^{\rho\gamma} x_\gamma = [g_{\nu\mu} \Lambda^\mu_\rho g^{\rho\gamma}] x_\gamma \\ &= \Lambda_\nu{}^\gamma x_\gamma = x_\gamma (\Lambda^{-1})^\gamma_\nu \end{aligned}$$

The reversed index structure appears naturally here.

Quick recap: Defined Lorentz transformation

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$$

which satisfies $(\Lambda^T)_{\rho}^{\mu} g_{\mu\nu} \Lambda^{\nu}_{\sigma} = g_{\rho\sigma}$

Covariant vectors transform under the inverse transformation:

$$x'_{\nu} = \Lambda_{\nu}^{\lambda} x_{\lambda} = x_{\lambda} (\Lambda^{-1})^{\lambda}_{\nu}$$

where we've defined

$$(\Lambda^{-1})^{\lambda}_{\nu} \equiv \Lambda_{\nu}^{\lambda} = g_{\nu\mu} g^{\lambda\rho} \Lambda^{\mu}_{\rho}$$

Note that Λ_{ν}^{λ} has the same index structure as Λ^T , the latter of which is given by

$$(\Lambda^T)_{\nu}^{\mu} = \Lambda^{\mu}_{\nu} \quad (\text{flipped order, not position})$$

The inverse transformations are:

$$x^{\mu} = (\Lambda^{-1})^{\mu}_{\nu} x'^{\nu}, \quad x_{\mu} = x'_{\nu} \Lambda^{\nu}_{\mu} \\ = (\Lambda^{-1})^{\nu}_{\mu} x'_{\nu}$$

The inverse metric can be used to write the unprimed coordinate vectors in terms of the primed ones (that's the part of an inverse)!

$$x^m = (\Lambda^{-1})^m_{\nu} x'^{\nu}$$

$$x_m = x'_{\nu} \Lambda^{\nu}_m$$

Now we'll consider the explicit form of some particular transformations.

Spatial rotations

An obvious subgroup of the set of Lorentz transformations is the set of rotations of the three spatial coordinates.

E.g., a rotation about the $z = x^3$ axis mixes the $x^1 = x$ and $x^2 = y$ coordinates, so it has the form:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \Lambda(R) \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

To derive a more explicit form, use the orthogonality relation: $\Lambda^T g \Lambda = g$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & a & c & 0 \\ 0 & b & d & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -(a^2+c^2) & -(ab+cd) & 0 \\ 0 & -(ab+cd) & -(b^2+d^2) & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \underset{\text{demand}}{=} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

So we have the requirements:

$$a^2 + c^2 = 1, \quad b^2 + d^2 = 1, \quad ab + cd = 0$$

These are easily satisfied by choosing

$$a = \cos\theta, \quad b = -\sin\theta, \quad c = \sin\theta, \quad d = \cos\theta$$

So the Lorentz transformation for this particular spatial rotation has the form:

$$\Lambda^\mu{}_\nu(R(\theta)) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Boosts: Next consider a boost along the $x' = x$ direction, which mixes the time x^0 & x' coords:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \Lambda(B) \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

$\Lambda(B)$ must satisfy: $\Lambda^T g \Lambda = g$:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} a^2 - c^2 & ab - cd \\ ab - cd & b^2 - d^2 \end{pmatrix}$$

So we require: $a^2 - c^2 = 1$, $b^2 - d^2 = -1$, $ab + cd = 0$

$$\Rightarrow a = \cosh \eta, b = -\sinh \eta, c = \sinh \eta, d = \cosh \eta$$

where η is the rapidity, $-\infty \leq \eta \leq \infty$.

A boost relates an inertial frame O' moving at constant velocity v ~~along the x direction~~ moving along the x' direction to an inertial frame.

We can solve for the velocity v by setting $x' = 0$:

$$x' = (-\sinh \eta)(ct) + (\cosh \eta)x = 0$$

From the original frame, the velocity is $v = x/t$, w/ x the origin of O' !

Defining $\beta = v/c$, we solve the equation above to find:

$$\beta = \frac{v}{c} = \frac{x}{ct} = \frac{\sinh \eta}{\cosh \eta} = \tanh \eta$$

Now,

$$\sinh \eta = \cosh \eta \cdot \beta$$

$$\Rightarrow \sinh^2 \eta = \cosh^2 \eta \cdot \beta^2$$

Then,

$$1 = \cosh^2 \eta - \sinh^2 \eta = \cosh^2 \eta (1 - \beta^2)$$

which is solved for

$$\cosh^2 \eta = \frac{1}{1 - \beta^2} \equiv \gamma^2$$

With γ the usual Lorentz boost factor:

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}, \quad \cosh \eta = \gamma, \quad \sinh \eta = \beta \cdot \gamma$$

A Lorentz boost thus takes the form:

$$\Lambda^\mu{}_\nu(B(\beta_x)) = \begin{pmatrix} \gamma & -\beta_x \gamma & 0 & 0 \\ -\beta_x \gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cosh \eta & -\sinh \eta & 0 & 0 \\ -\sinh \eta & \cosh \eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

These transformations lead to familiar concepts such as length contraction and time dilation.

E.g., for two events separated by proper time $\tau = t_2 - t_1$ in an inertial frame O , the time in a boosted frame O' is given by:

$$\tau' = t'_2 - t'_1 = \gamma(t_2 - \frac{\beta}{c} x_2) - \gamma(t_1 - \frac{\beta}{c} x_1)$$

$$= \gamma(t_2 - t_1) - \gamma \frac{\beta}{c} (x_2 - x_1)$$

(assuming they happen in same location in O)

$$= \gamma \tau$$

Discrete Spacetime Transformations

Rotations and boosts are examples of
 "proper orthochronous" Lorentz transformations
 $\rightarrow$ they are continuously connected to the identity.
 (take θ , β or $\eta \rightarrow 0$)

They also have positive determinant: $\det \Lambda = +1$
 and $\Lambda^0_0 \geq 1$.

$\rightarrow$ elements of $SO(3,1)^+$ ~~Elements~~

There are two particular transformations that
 are not continuously connected to the identity:

Parity: (space inversion)

$$P: \begin{array}{l} t \rightarrow t \\ \vec{x} \rightarrow -\vec{x} \end{array}, \quad \Lambda^\mu_\nu(P) = \text{diag}(1, -1, -1, -1)$$

Time-reversal:

$$T: \begin{array}{l} t \rightarrow -t \\ \vec{x} \rightarrow \vec{x} \end{array}, \quad \Lambda^\mu_\nu(T) = \text{diag}(-1, 1, 1, 1)$$

$\Rightarrow$ Both satisfy $\Lambda^T g \Lambda = g$.

A general Lorentz transformation can be decomposed into a ~~per~~ product of a pure rotation & a boost, with or without a factor of P or T .

There is another useful discrete transformation C which roughly interchanges particles & their antiparticles. However, C is not a spacetime symmetry (it is an internal symmetry acting on the particles/fields).

It is very useful to study properties of a system or a reaction (e.g., decay, scattering) under C , P & T and their combinations. We'll have a lot more to say on this later.

For a long time, it was thought P & T (or CP) were exact symmetries of nature. One of the most significant discoveries (by C.S. Wu) was the discovery that parity is violated by the weak interactions. The same is true for CP .

The combination CPT is an exact symmetry in all ^{xt} QFTs.

* Lorentz,
-invariant,
+ local, w/o
negative
norm states...

Spacetime interval & the light cone

Consider two events x^m , y^m & the difference

$$\Delta x^m = x^m - y^m = (c\Delta t, \Delta x, \Delta y, \Delta z)$$

The "interval" in Minkowski space is the same for all observers, and given by:

$$\begin{aligned}\Delta s^2 &= g_{\mu\nu} \Delta x^\mu \Delta x^\nu = (c\Delta t)^2 - (\Delta \vec{x})^2 \\ &= (x^m - y^m)^2\end{aligned}$$

Useful to visualize this on a light-cone diagram!

Any interval can be classified as:

$$- \Delta s^2 > 0$$

(timelike)

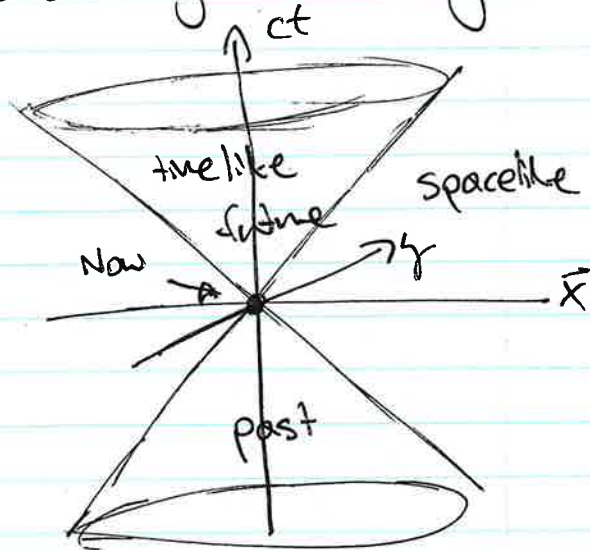
$$- \Delta s^2 < 0$$

(spacelike)

$$- \Delta s^2 = 0$$

(lightlike)

not causally connected.



Energy-Momentum Four Vector

Aside from $x^\mu = (ct, \vec{x})$ we can consider another important 4-vector: the energy-momentum 4-vector.

This will be extremely important for particle kinematics.

Consider the differential spacetime interval (line-element) of a particle:

$$ds^2 = c^2 dt^2 - |d\vec{x}|^2$$

In the particle's rest frame, $d\vec{x} = 0$, so $ds^2 = c^2 d\tau^2$ - the proper time.

In another frame (e.g., the "lab frame") where the particle is moving, we have:

$$\begin{aligned} c^2 d\tau^2 &= c^2 dt^2 - d\vec{x}^2 = c^2 dt^2 \left(1 - \frac{1}{c^2} \left(\frac{d\vec{x}}{dt} \right)^2 \right) \\ &= c^2 dt^2 (1 - \beta^2) = \frac{c^2 dt^2}{\gamma^2} \end{aligned}$$

or simply, $dt = \gamma d\tau$ (time dilation)

Now define the proper velocity of the particle:

$$\eta^\mu = \frac{dx^\mu}{d\tau}$$

The components of η^μ are:

$$\left. \begin{aligned} \eta^0 &= \frac{dx^0}{d\tau} = \gamma \frac{d}{dt}(ct) = \gamma c \\ \vec{\eta} &= \frac{d\vec{x}}{d\tau} = \gamma \frac{d\vec{x}}{dt} = \gamma \vec{v} \end{aligned} \right\} \eta^\mu = \gamma(c, \vec{v})$$

Note that the square of η^μ is

$$\eta^2 = \eta^\mu \eta_\mu = \gamma^2 c^2 - \gamma^2 v^2 = \gamma^2 c^2 (1 - \beta^2) = c^2$$

Now we can define the four-momentum as:

$$\begin{aligned} p^\mu &= m \eta^\mu = m \gamma (c, \vec{v}) \\ &= (\gamma m c, \gamma m \vec{v}) = \left(\frac{E}{c}, \vec{p} \right) \end{aligned}$$

where m is the rest mass of the particle, and the total energy and three momentum are given by

$$E = \gamma m c^2, \quad \vec{p} = \gamma m \vec{v}$$

At rest, $\vec{v} = 0$, $\gamma = 1 \Rightarrow E_{\text{rest}} = m c^2$

The kinetic energy is

$$T = E - m c^2 = (\gamma - 1) m c^2$$

The square of the four-momentum is a Lorentz invariant:

$$p^2 = p^\mu p_\mu = \left(\frac{E}{c}\right)^2 - |\vec{p}|^2 = m^2 \eta^\mu \eta_\mu = mc^2$$

$$(\rightarrow E^2 = |\vec{p}|^2 c^2 + m^2 c^4)$$

That $p^2 = mc^2$ is invariant is the statement that the rest mass is invariant (we'll never discuss a "relativistic mass" — mass = rest mass)

For a massless particle, the generalization that yields the Einstein relation $E = |\vec{p}| \cdot c$ is

$$p^\mu = (|\vec{p}|, \vec{p})$$

The four-momentum transforms ~~like a 4-vector~~ under Lorentz transformations like any 4-vector:

$$p'^\mu = \Lambda^\mu_\nu p^\nu$$

e.g., under a boost:

$$E' = \gamma(E - v p_x) \quad , \quad p'_x = \gamma(p_x - \frac{vE}{c^2})$$

Note that starting from a rest frame, boosting to a lab frame, we have $p_x = 0$, so

$$E' = \gamma E = \gamma mc^2$$

Thus, if we know a particle's total energy in the lab frame (common to measure in experiments) then we can immediately infer the boost as $\gamma = E_{\text{lab}} / mc^2$

For example;

Cosmic ray muons have mean energy $\sim$ few GeV

$$\rightarrow \gamma = (5 \text{ GeV}) / m_\mu c^2$$

$$= (5 / 0.104) \approx 50$$

thus their lifetime in our ("lab" frame) is

$$\tau_{\text{lab}} = \gamma \tau_\mu \approx 50 \times 2.2 \times 10^{-6} \text{ s}$$

$$\approx 1 \times 10^{-4} \text{ sec (long enough to reach the surface)}$$

$$(\Rightarrow c \tau_{\text{lab}} \approx 3 \times 10^4 \text{ meters})$$

Another example: the B^0 mesons have lifetimes
 $\sim 1.5 \times 10^{-12}$ s ($c\tau = 0.045$ cm)

their mass is ~ 5280 MeV (# of feet in a mile!)

Frequently produced at the LHC w/ energies
 ~ 50 GeV $\Rightarrow \gamma \sim 10$.

$\Rightarrow \gamma c\tau_B \sim 5$ mm — long enough to see displacement in a detector!

This is important for "b-tagging" jets at the LHC, a capability that's crucial for Higgs precision studies, for example.

Kinematics of collisions: fixed target vs. colliders

It's also useful to consider the kinematics of two types of scattering experiments:

- i) fixed target experiment:
(e.g., proton incident on chunk of lead)
made of nucleons
- ii) ~~pro~~ colliders: protons incident on other protons moving opposite directions.

Fixed target: particle A impacts particle B, w/ B at rest in the lab system.

In the lab frame:

$$P_A^\mu = \left(\frac{E_A}{c}, \vec{p}_A \right) \quad P_B^\mu = (M_B c, \vec{0})$$

$$E_A = \sqrt{|\vec{p}_A|^2 c^2 + M_A^2 c^4}$$

The total energy-momentum squared is:

$$P^2 = (P_A + P_B)^2 = (P_A + P_B)^\mu (P_A + P_B)_\mu$$

$$= \left(\frac{E_A}{c} + M_B c \right)^2 - (\vec{p}_A + \vec{0})^2$$

$$= \frac{E_A^2}{c^2} + 2M_B E_A + M_B^2 c^2 - |\vec{p}_A|^2$$

$$\text{or, } p^2 = m_A^2 c^2 + m_B^2 c^2 + 2m_B E_A$$

In the center of mass (CM) system, defined by the condition that the total spatial momentum vanishes:

$$\vec{p}_A^* + \vec{p}_B^* = 0$$

The square of the total four momentum is:

$$(p^*)^2 = (p_A^* + p_B^*)^2 = (E_A^* + E_B^*, 0)^2 \equiv E_*^2$$

where we've defined $E^* = E_A^* = E_B^*$ as the total CM energy.

Since p^2 is a Lorentz invariant, we have for the fixed target configuration:

$$E^* = \sqrt{(p^*)^2} = \sqrt{p^2} = \sqrt{m_A^2 c^2 + m_B^2 c^2 + 2m_B E_A}$$

Typically, the beam energy is much larger than the rest energy of the particles: $E_A \gg m_A c^2, m_B c^2$

$$\Rightarrow E^* \simeq \sqrt{2m_B E_A}$$

→ This is much ~~less~~ smaller than the total lab frame energy, as some of the total energy is ~~not available~~ part of the relative motion.

To e.g., create a particle, we need enough CM energy: $E^* > M c^2$

In a collider configuration, where A and B are moving head on, things are much better.

If E_A & E_B are the same (as at most, though not all colliders) then:

$$E^* = E_A + E_B = 2E$$

→ this is much more favorable to get to high energies & produce heavy particles.

(Of course, this is very hard as the beams must be focused carefully on to each other — this is part of the challenge in reaching high luminosity.)

Other four-vectors:

Another operator we'll encounter frequently is the 4-gradient ∂_m :

$$\partial_m \equiv \frac{\partial}{\partial x^m} = \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3} \right) = \left(\frac{1}{c} \frac{\partial}{\partial t}, \vec{\nabla} \right)$$

$$= g_{mv} \partial^v$$

$$\partial^v = \frac{\partial}{\partial x_m} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\vec{\nabla} \right) = g^{mv} \partial_v$$

These transform ~~as~~ as contra/co-variant 4-vectors appropriate to their index structure.

We can also define the d'Alembertian operator:

$$\square \equiv g_{mv} \partial^m \partial^v = \partial^m \partial_m = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2$$

which is a scalar (invariant) under Lorentz transformations.

A final example that we'll come back to repeatedly later on: Electromagnetism.

Classical Electromagnetism

Maxwell's equations can be written:

$$(i) \quad \vec{\nabla} \cdot \vec{E} = \rho \quad \leftarrow \text{charge density}$$

$$(ii) \quad \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

$$(iii) \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$(iv) \quad \vec{\nabla} \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \vec{j} \quad \leftarrow \text{current density}$$

Relativistic invariant, but not obvious in this form.
Can also be written using the scalar/vector potentials:

$$\vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

The potentials can be packaged into a four-vector:

$$A^\mu = (\phi, \vec{A})$$

Then define the "field strength tensor":

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

In components:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E^x & -E^y & -E^z \\ E^x & 0 & -B^z & B^y \\ E^y & B^z & 0 & -B^x \\ E^z & -B^y & B^x & 0 \end{pmatrix}$$

or,

$$F^{0i} = -E^i, \quad F^{ij} = -\epsilon^{ijk} B^k$$

Note that $F^{\mu\nu} = -F^{\nu\mu}$.

Further we define the 4-current density:

$$j^{\mu} = (\rho, \vec{j})$$

Then, the "inhomogeneous" Maxwell eqns. (i) & (iv) can be written:

$$\partial_{\mu} F^{\mu\nu} = j^{\nu}$$

While (ii) & (iii) follow from the antisymmetry of $F^{\mu\nu}$, which leads to the Bianchi-identity

$$\partial^{\mu} F^{\nu\rho} + \partial^{\nu} F^{\rho\mu} + \partial^{\rho} F^{\mu\nu} = 0$$

~~The E & B~~ The vector A^{μ} transforms as a Lorentz vector, while $F^{\mu\nu}$ as a rank-2 tensor.